

Citations

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CuPV-1 Alignment Reanalysis

PIPELINE TOOLS

NCBI ORF Finder

Identifies open reading frames (ORFs) in the viral genome sequence.

T-Coffee

Multiple sequence alignment tool, instead of ClustalW for aligning the sequences.

MEGA (Molecular Evolutionary Genetics Analysis) Software

Performs phylogenetic analysis using Maximum Likelihood (ML) methods to build evolutionary trees.

NCBI GenBank

Source of reference sequences for comparison in the analysis.

Nextflow

Workflow management tool to automate the pipeline and manage computational tasks.

Biopython

Python library for bioinformatics tasks, useful for handling sequence data and potentially managing data within the pipeline.

BLAST (Basic Local Alignment Search Tool)

Verifies sequence similarity and identities within the dataset.

BIOL - 3951 PROJECT
Claire Gallant
Abdelrahman Conber
Haley Leonard

Manual



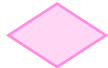
Automated



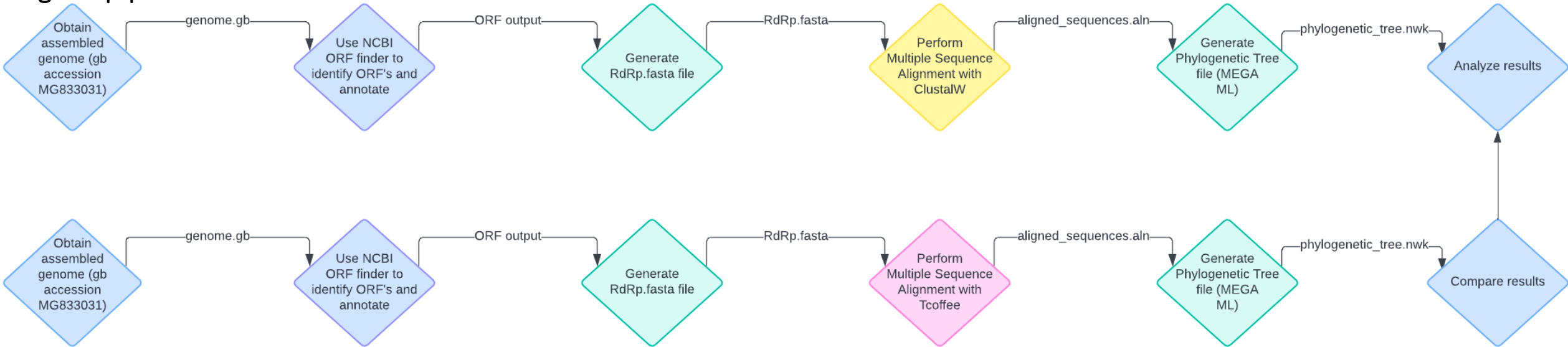
Original



Our changes



Original pipeline



Our pipeline